Matthew Au-Yeung

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Education

University of Waterloo *BCS in Computer Science* Sept 2022 – Apr 2027

- GPA: 3.8 (transcript)
- **Coursework:** Data Structures & Algorithms, Operating Systems, Distributed Systems, Concurrency, Compilers, Databases, Data-Intensive Applications, Cryptography & Security

Experience

Mechanical Orchard May 2025 – Present (3 terms)

Software Engineer Intern *Remote*

- Implemented a **COBOL compiler** and custom **bytecode VM** in Java, with a native database engine and concurrency support in the runtime, compiling **20,000+ programs**
- Engineered a **concolic tester** shadowing the bytecode symbolically with **CVC5**, deriving test suites covering **7.6M of 10M** client production lines in hours to validate Java rewrites
- Developed an **LLM agent** pairing COBOL type and call-graph analysis with a generate-compile-repair loop, shipping **20+ Java rewrites** to client production at **100%** match on a year of captured data

Fab2 (Atomic Semi) May 2026 – Aug 2026

Software Engineer Intern *San Francisco, CA*

- Built a Go gRPC **deployment server** and Rust **host updater** delivering versioned builds in S3 to **100+ tools** and recording who deployed what, cutting **hours** of deploy checks with a protocol each tool implements
- Sandboxed third-party services with **systemd** and **nftables**, forcing all traffic through a Go **egress proxy** that manages each service's DNS, HTTPS tunnels, and Postgres connections
- Self-hosted Claude Gateway and Bifrost behind a **Go service** handling auth, provider routing, and usage accounting for **100+ employees** on export-compliant Vertex AI

Ford Motor Company Jan 2025 – Apr 2025

Software Developer Intern *Waterloo, ON*

- Created a **Kotlin Android service** tracing **100% of Apple CarPlay buttons** for **5 vehicle UX teams**
- Increased mobile projection test coverage from **70% to 83%** using Coroutines, JUnit, and Robolectric

Township of Centre Wellington May 2024 – Aug 2024

Software Developer Intern *Fergus, ON*

- Shipped a document portal and shared notes tracking **1K+ locations** for **100+ officials** using C# ASP.NET Core, SignalR WebSockets, TypeScript, and MSSQL

Research

University of Waterloo Sept 2025 – Apr 2026

Compilers Research Assistant *Waterloo, ON*

- Supervised by Dr. Peter Buhr: Introduced new global linking semantics to the **Cforall** compiler (a concurrent extension of C), and eliminated deadlocks in its threading runtime

University of Waterloo May 2024 – Jan 2025

Compilers Testing Research Assistant *Waterloo, ON*

- Supervised by Dr. Chengnian Sun: Built a MIPS assembly interpreter and graphical debugger in Kotlin used annually by **1,000+ students** to debug compiler outputs

Projects

Distributed Go Channels — *Go, gRPC, hashicorp/raft* m4tth3/dchan

- Created a Go library providing **native channels** for many-to-many distributed messaging over Raft and gRPC
- Replaced per-message dialing with a gRPC-based Raft transport that shares long-lived connections

Skills

Languages: Go, Rust, C/C++, Java, Kotlin, Python, TypeScript, C#, Elixir
Tools and Frameworks: gRPC, Protobuf, Raft, Spring, ASP.NET Core, Nix
Databases and Cloud: S3, PostgreSQL, Microsoft SQL Server (MSSQL), GCP, Docker, GitHub Actions